European Heart Journal - Quality of Care and Clinical Outcomes (2026) **12**, 805–815
https://doi.org/10.1093/ehjqcco/qcaf156

ORIGINAL ARTICLE

Mitral valve prolapse: incidence and prognosis related to serum levels of sex hormone-binding globulin

Zhaolin Fu[1,†], Zhennan Lin[2,3,†], Qiang Li[1,†], Xunan Guo[1], Xinmin Liu[1], Kaiyang Lin[2,3,*], and Guangyuan Song [1,*]

[1]Interventional Center of Valvular Heart Disease, Beijing Anzhen Hospital Affiliated to Capital Medical University, Beijing 100029, China; [2]Department of Cardiology, Shengli Clinical Medical College of Fujian Medical University, Fujian Provincial Hospital, Fuzhou University Affiliated Provincial Hospital, Fuzhou 350000, China; and [3]Fujian Provincial Key Laboratory of Cardiovascular Disease, Fujian Provincial Center for Geriatrics, Fujian Clinical Medical Research Center for Cardiovascular Diseases, Fujian Cardiovascular Institute, Fuzhou 350000, China

Received 5 August 2025; revised 26 October 2025; accepted 8 December 2025; online publish-ahead-of-print 16 December 2025

Aims

Mitral valve prolapse (MVP) is common, with higher prevalence in women. We hypothesized that sex hormones and regulatory protein sex hormone binding globulin (SHBG) play a pivotal role in development and prognosis of MVP.

Methods and results

We prospectively study 472 178 participants in the UK Biobank. Univariable and multivariable regression analyses estimated associations of SHBG and sex hormones (including testosterone and oestradiol) with incidence and prognosis of MVP, and differences between sexes. Mendelian randomization (MR) further assessed the causal relationship between SHBG and MVP. Out of 472 178 participants, 970 patients without a baseline MVP diagnosis developed MVP over a median follow-up of 13.67 years. Serum SHBG and testosterone were positively associated with MVP incidence. In the multivariable model adjusted for confounders, SHBG remained significantly associated with MVP risk [hazard ratio (HR) = 1.440, 95% confidence interval (CI): 1.201–1.726]. Compared to individuals with the lowest quintile of SHBG level, those in the second, third, and fourth quintiles had an 8.1, 39.1, and 49.3% higher risk of incident MVP. The association between SHBG and incident MVP was consistent across sexes (*P* for interaction = 0.4434). Elevated SHBG level was also significantly associated with all-cause mortality (HR = 2.223, 95% CI: 1.430–3.457) in MVP patients. One-sample and two-sample MR using European ancestry GWAS data supported a causal relationship between genetically predicted SHBG levels and increased MVP risk in both females and males.

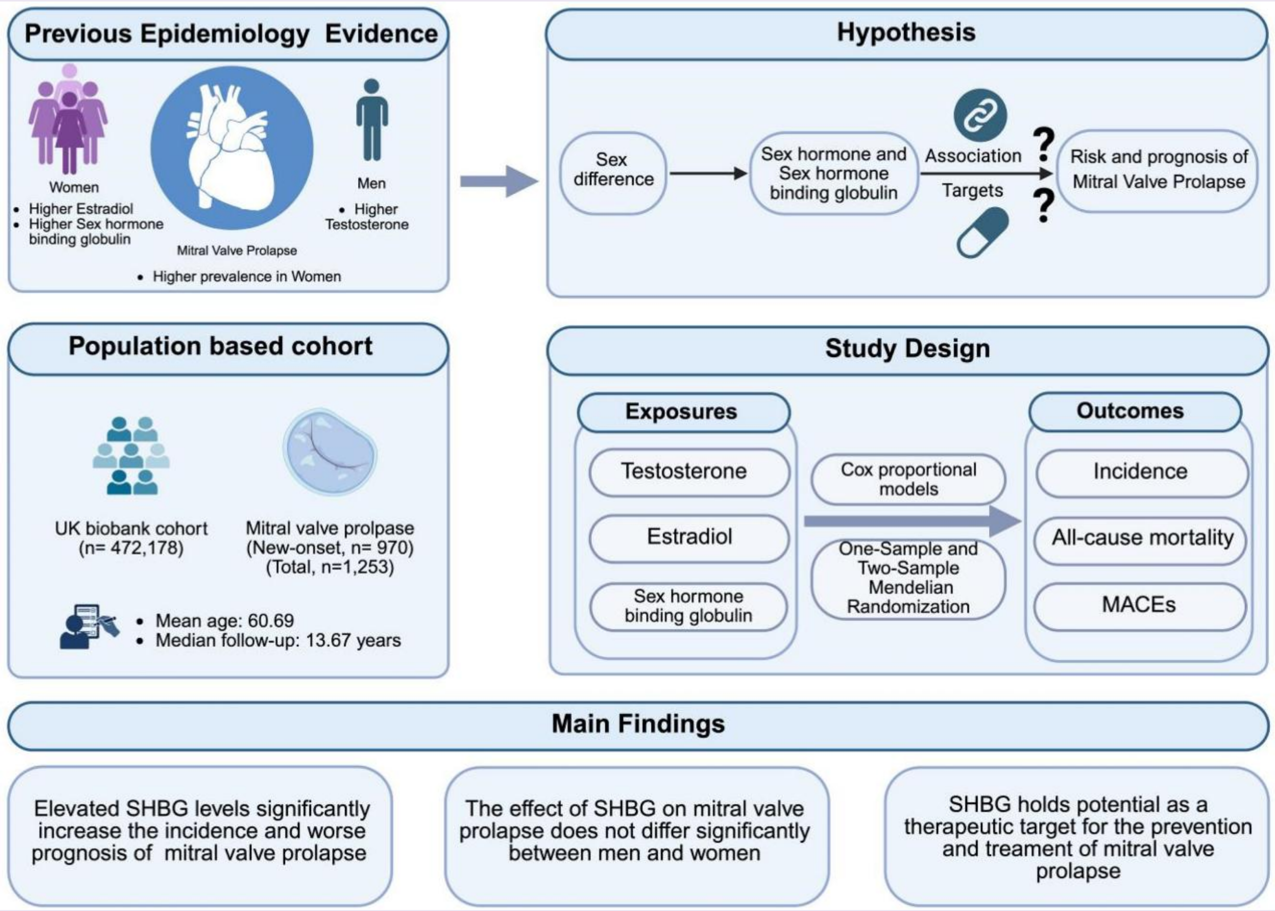
Conclusion

Elevated serum SHBG levels are associated with increased risk and adverse prognosis of MVP, providing insights into sex-related differences, risk stratification, and potential therapeutic targets.

*Corresponding authors. Tel: (+86) 13801120805, Email: songgy_anzhen@vip.163.com; Tel: (+86) 13559355708, Email: lky7411@sina.com

[†]The first three authors contributed equally to this work.

Graphical abstract



Keywords Mitral valve prolapse • Sex hormone-binding globulin • Sex hormones • Mendelian randomization

Key Learning Points

What is already known

- Mitral valve prolapse (MVP) is a common valvular heart disease with sex differences in incidence.
- Serum levels of sex hormones, including testosterone and oestradiol, as well as sex hormone-binding globulin (SHBG), a key regulator of the bioavailability of sex hormones, differ significantly between sexes, with SHBG levels being substantially higher in females than in males.
- Testosterone, oestradiol, and SHBG are associated with the development of cardiovascular diseases and contribute to sex differences in cardiovascular risk.

What this study adds

- Serum levels of SHBG is a risk factor for the incidence and poor prognosis of mitral valve prolapse in both sexes.
- Serum levels of testosterone and oestradiol are not associated with the incidence and prognosis of MVP.
- SHBG may contribute to the sex differences in MVP prevalence and may be a potential target for therapeutic interventions.

Introduction

Mitral valve prolapse (MVP) is one of the most common valvular heart diseases, affecting 2–3% of the community-based sample population.[1-3] Over 10% of individuals with MVP progress to severe mitral regurgitation, with a complication rate as high as 10–15%, including heart failure, atrial fibrillation, life-threatening ventricular arrhythmias, and cardiovascular death. The incidence of MVP is different between sexes, with women having a higher risk of MVP and tending to present at an earlier age compared to men.[4-9] Furthermore, a meta-analysis suggested that the incidence of MVP is highest during adolescence, a developmental stage marked by significant fluctuations in sex hormone levels,[10] which highlighted the need to investigate linkages between sex hormones and MVP.

Sex hormones, including testosterone and oestradiol, along with their key regulator, sex hormone-binding globulin (SHBG), have been shown to contribute to sex differences in some cardiovascular diseases.[11-14] Testosterone is the primary sex hormone in males, while oestradiol, the main biologically active form of oestrogen, is the primary sex hormone in females. However, testosterone and oestradiol are produced in both sexes, influencing various health outcomes. Sex hormone-binding globulin is a protein primarily synthesized by the liver and secreted into the bloodstream, where it tightly binds to sex hormones, regulating their bioavailability, but also directly signalling through specific cell surface receptors.[13] Sex hormone-binding globulin expression is itself influenced by sex hormones, with oestrogen promoting and testosterone inhibiting SHBG synthesis. For this reason, blood concentrations of SHBG are higher in females than in males. A previous study has demonstrated that SHBG can bind to oestrogen receptors on the surface of lymphocytes, promoting oestradiol uptake and subsequently intracellular signal transduction.[15] The activation of oestrogen receptors on the membrane of T cells has been implicated in collagen degradation in the heart valve, potentially contributing to sex-specific predisposition in rheumatic heart valve disease.[16] Based on these findings, it is speculated that sex hormones and SHBG may play a role in the pathogenesis of MVP.

To test this hypothesis, we utilized the prospective cohort of the UK Biobank to investigate the relationships of SHBG and sex hormones (including testosterone and oestradiol) with the incidence and prognosis of MVP, and to evaluate whether these relationships differed by sex. We also applied Mendelian randomization analysis to confirm the causality between SHBG concentrations and MVP in both males and females (*Graphical Abstract*).

Methods

Study design and population

The study design of UK Biobank has been described in detail elsewhere (http://www.ukbiobank.ac.uk/).[17] Briefly, the source population was adults aged 40–69 years who lived ≤40 km of 22 study assessment centres situated throughout England, Wales, and Scotland. Approximately 9.2 million people registered in the National Health Service were mailed invitations to participate in the study. Among these people, 503 317 (5.5%) visited the assessment centres in 2006–10 and were enrolled.[18] Participants were given touchscreen questionnaires and physical examinations, and provided biological samples for molecular/genetic analyses. Our datasets included 502 143 participants (application number: 105 322). After excluding participants who were not of white-European ancestry and participants who already had MVP at baseline, 472 178 participants were included for the analysis. The UK biobank has received approval by the North West Multi-Centre Research Ethics Committee. The detailed study design was presented in Supplementary material online, *Figure S1*.

Definition of outcomes

For the analysis of the incidence, the primary outcome was the new occurrence of MVP. We defined MVP using in-patient hospital diagnoses coded according to the ICD-10 (I34.1) classifications, which were obtained from primary care or hospital admission sources. The diagnostic validity of valvular heart disease using ICD-10 codes in national registers demonstrated high accuracy (positive predictive value > 80%), with most cases showing moderate to severe severity.[19-21] Individuals with a documented diagnosis of MVP earlier than the date of attending assessment centres were excluded. For the analysis of prognosis, outcomes of MVP patients included all-cause mortality, and major cardiovascular events (MACEs). Major cardiovascular events included fatal or non-fatal cardiovascular events (heart failure, myocardial infarction, ischaemic stroke, and intracerebral haemorrhage) during the follow-up after inclusion in the study, which were also defined using ICD-10 diagnostic codes (see Supplementary material online, *Table S1*). Participants were followed from enrolment until death, loss to follow-up, or 31 October 2022, whichever came first.

Serum sex hormone-binding globulin and sex hormone measurements

Blood biospecimens were collected at the time of enrolment in serum separator tubes, transported at 4°C to the central UK Biobank processing laboratory, aliquoted, and stored at −80°C.[22] Serum testosterone (nmol/L) was analysed using a one-step competitive immunoassay, oestradiol (pmol/L) using a two-step competitive immunoassay, and SHBG (nmol/L) using a two-step sandwich immunoassay. These hormones were measured using a Beckman Coulter Unicel DxI 800 immunoassay analyzer. The lower limits of detection for testosterone, oestradiol, and SHBG were <0.35 nmol/L, <175 pmol/L, and <0.33 nmol/L, respectively. Full information on serum biochemistry and quality control procedures can be found at the UK Biobank website. The UK Biobank did not distinguish between values above and below the limits of detection; therefore, missing values for these reasons could not be imputed.

Covariates

Demographic characteristics, lifestyle factors, and other confounders collected at baseline for each participant were included in our models as covariates, including age, sex, smoking status, drinking status, education, physical activity, body mass index (BMI), and presence of medical conditions at baseline [diabetes (E10-E14), hypertension (I10), and dyslipidaemia (E78.0-E78.5)]. Participants were grouped into current smokers and non-current smokers, as well as current drinkers and non-current drinkers, according to their smoking and drinking status at baseline. Education levels were classified as high level (college or university degree), intermediate level (Advanced levels/Advanced Subsidiary levels, Ordinary levels/General Certificate of Secondary Education, or equivalent), and low level (others), according to their qualification. Physical activity was defined as ideal status of ≥150 min/week of moderate exercise or ≥75 min/week of vigorous exercise or 150 min/week of mixed activity. Body mass index was calculated as weight in kilograms divided by height in metres squared. The presence of diabetes, hypertension, and dyslipidaemia were determined based on the record of hospital inpatient visits. Individuals whose first inpatient diagnosis occurred before the date of their visit to assessment centres were classified as having these conditions at baseline. Additionally, responses to questions regarding 'Vascular/heart problems diagnosed by a doctor', 'Diabetes diagnosed by doctor', and 'Medication for cholesterol, blood pressure, diabetes' (data fields: 6150, 2444, 6153, and 6177) were also considered to supplement their medical conditions. Since blood samples from these participants were not generally collected in a standardized fasting state, we did not use lipid and glucose levels to assess their metabolic profiles.

Association analyses

Continuous variables were presented as mean ± standard deviation or median [interquartile ranges (IQR)], and categorical variables were presented as numbers (percentage). The differences in characteristics between two groups were tested using the Student's *t*-test, Wilcoxon rank-sum test, and χ^2 test for normally distributed, non-normally distributed, and categorical variables, respectively. The serum concentrations of SHBG and sex hormones were log-transformed before analysis due to their skewed distribution. Both the continuous and quantile forms of SHBG and sex hormones were used to evaluate their associations with MVP, using univariable and multivariable Cox proportional hazards models. Hazard ratios (HRs) and 95% confidence intervals (CIs) were estimated. Covariates in the multivariable-adjusted model included age, sex, education, smoking status, drinking status, physical activity, BMI, diabetes, hypertension, and dyslipidaemia. The dose-response relationships of SHBG and sex hormones with incident MVP were examined using restricted cubic splines with three manual knots. To investigate whether the associations differed by sex, subgroup analysis was conducted and the interaction effect was evaluated. Additionally, other subgroup analyses stratified by age, obesity, physical activity, diabetes, hypertension, dyslipidaemia, smoking, and drinking were also conducted.

Mendelian randomization analyses

The design and core assumptions of MR was presented in Supplementary material online, *Figure S2* as directed acrylic graphs. For two-sample MR, we used the summary statistics of the largest genome-wide association study (GWAS) for SHBG, conducted among 368 929 UK Biobank participants, to select instrumental variables.[23] The associations between instrumental variables and MVP were obtained from the summary statistics of the largest meta-GWAS for MVP (including 439 533 European participants in six cohorts).[24] We also selected sex-specific instrumental variables from sex-stratified GWASs for SHBG (214 989 females and 185 221 males in the UK Biobank study) to evaluate sex-specific causalities. The information on these GWASs is detailed in Supplementary material online, *Table S2*. Single nucleotide polymorphisms (SNPs) were screened as instrumental variables based on the following conditions: (i) keeping SNPs significantly associated with exposure ($P < 5 \times 10^{-8}$); (ii) excluding SNPs with an F-statistic <10 to avoid weak instrument bias; (iii) keeping independent SNPs that were identified using linkage disequilibrium clumping ($r^2 < 0.001$ within a window of 10 Mb) based on the European 1000 genomes panel. All instrumental variables are listed in Supplementary material online, *Table S3*. Causal effects were estimated based on the variants-specific Wald ratio method. When two or more variants were used, the inverse variance weighting with multiplicative random effect method was applied to meta-analyse the Wald estimates. The causal effects were reported as odds ratios (ORs) and 95% CIs per unit increase in SHBG. Several methods were used to test the robustness of MR estimates, i.e. Weighted median, MR-Egger, and MR-PRESSO methods. Cochran Q test was used to test heterogeneity,[25] and MR-Egger's intercept was used to estimate the horizontal pleiotropy.[26] We also performed one-sample MR analysis in the UK Biobank population. Those SNPs that were screened from the GWAS of the UK Biobank were used, and their effect sizes were obtained from the GWAS of deCODE.[27] A polygenic risk score for SHBG was calculated and used as instrumental variable. The two-stage least square regression was applied to estimate the causal relationship of SHBG with MVP. The covariates included age, sex, education, smoking status, drinking status, physical activity, BMI, diabetes, hypertension, and dyslipidaemia.

All analyses were performed using the R software (R version 4.0.5; R Foundation for Statistical Computing, Vienna, Austria). All *P*-values are two-sided, and $P < 0.05$ was set to demonstrate statistical significance.

Sensitivity analyses

To further account for potential confounding factors in the association between SHBG and the risk and prognosis of MVP, additional adjustments were made for serum total protein levels, mitral regurgitation, occurrence of mitral valve repair surgery, scoliosis, Marfan syndrome, and the use of medication including beta-blocker, angiotensin converting enzyme inhibitor (ACEI), angiotensin II receptor blocker (ARB), and calcium channel blocker (CCB).

In the MR analysis, we conducted several sensitivity analyses to test the robustness: (i) due to the potential sample overlap between the SHBG GWASs and MVP GWAS, the MRlap method was employed, enabling the correction for potential sample overlap by cross-trait linkage disequilibrium-score regression.[28] (ii) We performed two additional two-sample MR analyses based on GWAS data for SHBG from the deCODE study (35 559 participants from Iceland).[27] First, we used instrumental variables with a $P < 1 \times 10^{-6}$ that were selected from the GWAS of the deCODE study (see Supplementary material online, *Table S4*). Second, we used the instrumental variables selected from the GWAS of the UK Biobank and replaced their effect sizes and standard errors with those from the deCODE study (see Supplementary material online, *Table S5*). (iii) Previous studies have identified causal relationships of BMI, Marfan syndrome, and scoliosis with MVP.[13,29-32] To exclude the potential confounding of these variables (see Supplementary material online, *Table S3*), we removed SNPs significantly associated with them ($P < 5 \times 10^{-8}$) that were documented in the Phenoscanner[33] or the GWAS Catalog[34] (https://www.ebi.ac.uk/gwas/). Then, revised two-sample MR analyses were performed.

Results

Baseline characteristics of the study population

Among the 472 178 participants included in the study, the average age was 57.30 ± 8.02 years, and 45.5% of participants were male. At baseline, the average serum levels of SHBG, testosterone, and oestradiol were 45.63 (IQR: 32.74–64.33) nmol/L, 3.73 (IQR: 1.01–11.62) nmol/L, and 0.31 (IQR: 0.22–0.53) nmol/L, respectively. After a median follow-up of 13.67 years, 970 incident MVP cases were identified. Compared with non-MVP participants, the MVP patients were on average older (60.7 ± 7.01 vs. 57.3 ± 8.02 years, $P < 0.0001$) and had a higher proportion of males (54.9% vs. 45.5%, $P < 0.0001$). We considered that the higher prevalence of MVP among males than females may be due to the fact that the UK Biobank study population comprises individuals primarily aged 40 years and above. Moreover, participants who developed MVP had a higher serum level of SHBG [51.01 (IQR: 37.31–67.89) vs. 45.63 (IQR: 32.74–64.33) nmol/L] and were more likely to be current smokers, with a lower BMI, and have higher prevalence of diabetes, and hypertension, and dyslipidaemia at baseline (*Table 1*).

Associations of sex hormone binding globulin and sex hormones with mitral valve prolapse incidence

At univariable analysis, serum SHBG (HR = 1.524, 95% CI: 1.331, 1.745) and testosterone (HR = 1.196, 95% CI: 1.131, 1.263) were positively associated with incident MVP. After adjusting for multiple covariates, we found that with 1 unit increase in the log-transformed concentration of SHBG, the risk of incident MVP increased by 44.0% (HR = 1.440, 95% CI: 1.201, 1.726). Conversely, there was no significant association between sex hormones and MVP (*Table 2*). When compared to individuals with the first quintile of SHBG, those with the second, third, and fourth quintiles of SHBG had 8.1% (HR = 1.081, 95% CI: 0.860, 1.358), 39.1% (HR = 1.391, 95% CI: 1.111, 1.741), and 49.3% (HR = 1.493, 95% CI: 1.166, 1.913) higher risk of incident MVP, respectively (*P* for trend < 0.0002) (*Figure 1*). Dose-response curves demonstrated that SHBG was linearly associated with MVP (*P* for nonlinear = 0.917 and *P* for linear = 0.0004) (see Supplementary material online, *Figure S3*).

We further evaluate the differences in the effects of sex hormones and SHBG on incident MVP across by sexes. We observed that, compared with males, the average levels of SHBG and oestradiol were higher and the average level of testosterone was lower (*Figure 2A* and Supplementary material online, *Table S6*). We then observed that SHBG was positively associated with incident MVP in both males and females, and the effects of SHBG were not different (*P* for interaction = 0.4426) (*Figure 2B*).

In addition, subgroup analyses for other confounders showed no significant interaction effects of SHBG with age, obesity, hypertension, diabetes, dyslipidaemia, current smoking, current drinking, and ideal physical activity (all *P* for interaction > 0.05) (see Supplementary material online, *Table S7*).

In the sensitivity analyses, after adjusting for potential confounders, including serum total protein levels, mitral regurgitation, scoliosis, Marfan syndrome, and the use of medication including beta-blocker, ACEI, ARB, and CCB. Sex hormone-binding globulin remained significantly associated with MVP incidence (see Supplementary material online, *Table S8*).

Table 1 Baseline characteristics of the study population

Characteristics	All participants (*n* = 472 178)	Non-MVP (*n* = 471 208)	MVP (*n* = 970)	*P*
Age, years	57.30 ± 8.02	57.29 ± 8.02	60.70 ± 7.01	<0.0001
Male, *n* (%)	214 988 (45.53)	214 456 (45.51)	532 (54.85)	<0.0001
Current smoking, *n* (%)	49 462 (10.51)	49 394 (10.52)	68 (7.01)	0.0005
Current drinking, *n* (%)	439 887 (93.25)	438 977 (93.25)	910 (93.81)	0.5211
Education, *n* (%)				0.0459
Low	161 079 (34.68)	160 759 (34.69)	320 (33.44)	
Intermediate	153 474 (33.05)	153 181 (33.05)	293 (30.62)	
High	149 855 (32.27)	149 511 (32.26)	344 (35.95)	
Ideal physical activity, *n* (%)	240 572 (54.47)	240 038 (54.46)	534 (57.98)	0.035
BMI, kg/m^2	27.41 ± 4.78	27.41 ± 4.79	25.76 ± 3.84	<0.0001
SBP, mmHg	137.98 ± 18.66	137.98 ± 18.66	139.81 ± 18.63	0.0029
DBP, mmHg	82.17 ± 10.26	82.17 ± 10.26	80.70 ± 10.13	<0.0001
Hypertension, *n* (%)	138 092 (29.25)	137 758 (29.24)	334 (34.43)	0.0004
Triglycerides, mmol/L	1.49 (1.05, 2.15)	1.49 (1.05, 2.15)	1.41 (1.04, 1.96)	0.0018
Total cholesterol, mmol/L	5.71 ± 1.14	5.71 ± 1.14	5.59 ± 1.11	0.001
LDL-C, mmol/L	3.57 ± 0.87	3.57 ± 0.87	3.47 ± 0.83	0.0004
HDL-C, mmol/L	1.45 ± 0.38	1.45 ± 0.38	1.47 ± 0.38	0.1229
Dyslipidaemia, *n* (%)	83 399 (17.66)	83 207 (17.66)	192 (19.79)	0.0891
Glucose, mmol/L	4.93 (4.60, 5.31)	4.93 (4.60, 5.31)	4.89 (4.57, 5.26)	0.0255
Diabetes mellitus, *n* (%)	23 592 (5.00)	23 568 (5.00)	24 (2.47)	0.0004
Mitral regurgitation, *n* (%)	684 (0.14)	651 (0.14)	33 (3.40)	<0.0001
SHBG, nmol/L	45.63 (32.74, 64.33)	45.61 (32.73, 64.31)	51.01 (37.31, 67.89)	<0.0001
Testosterone, nmol/L	3.73 (1.01, 11.62)	3.67 (1.01, 11.62)	9.20 (1.11, 12.93)	<0.0001
Oestradiol, nmol/L[a]	0.31 (0.22, 0.53)	0.31 (0.22, 0.53)	0.25 (0.19, 0.48)	0.0100

Normally distributed variables are presented as mean ± standard deviation, and the differences were tested using Student's *t*-test. Variables with skewed distribution are presented as median (interquartile range), and the differences were tested using Wilcoxon rank-sum test. Categorical variables are presented as frequencies (proportions), and the differences were tested using χ^2 test. BMI, body mass index; DBP, diastolic blood pressure; HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; MVP, mitral valve prolapse; SBP, systolic blood pressure; SHBG, sex hormone-binding globulin.

[a]Only 70 696 participants were detected oestradiol.

Associations of sex hormone-binding globulin and sex hormones with mitral valve prolapse prognosis

A total of 1253 MVP patients were used for this analysis (including prevalent and incident MVP cases). The baseline characteristics of MVP patients with different prognoses were shown in Supplementary material online, *Table S9*. After a median follow-up of 6.87 years, 168 patients were documented as having all-cause mortality. After a median follow-up of 10.94 years, 598 MACE cases were identified. At univariable analysis, SHBG was associated with all-cause mortality, testosterone was associated with all-cause mortality and MACEs, and oestradiol was associated with MACE. In the multivariable-adjusted model, the association was only present for SHBG. With 1 unit increase in SHBG among MVP patients, their risk of all-cause death increased by 1.223-fold (HR = 2.223, 95% CI: 1.430, 3.457) (*Table 3*). In the sensitivity analyses, after adjusting for potential confounders, including serum total protein levels, mitral regurgitation, scoliosis, Marfan syndrome, occurrence of mitral valve repair surgery, and the use of medication including beta-blocker, ACEI, ARB, and CCB, SHBG remained significantly associated with all-cause mortality(see Supplementary material online, *Table S8*).

Causal relationship between sex hormone-binding globulin and mitral valve prolapse—Mendelian randomization inference

The two-sample MR analyses showed that an increase in genetically predicted SHBG was significantly related to a higher MVP risk in the total population (OR = 1.576, 95% CI: 1.170, 2.122), female (OR = 1.350, 95% CI: 1.155, 1.578), and male (OR = 1.286, 95% CI: 1.126, 1.469). No significant pleiotropy was presented (*Figure 3*). The one-sample MR supports a causal relationship between genetically predicted SHBG and a higher MVP risk in the total population (OR = 1.673, 95% CI, 1.150, 2.433), female (OR = 2.572, 95% CI: 1.067, 6.199), and male (OR = 2.331, 95% CI: 1.088, 4.995) (*Figure 3*).

In the sensitivity analyses, using the MRlap method to correct for potential sample overlap, the association between SHBG and MVP became more statistically significant in the overall population as well as in both males and females (see Supplementary material online, *Table S10*). Additionally, using GWAS for SHBG from the deCODE study for external validation, the causal association between SHBG and MVP remained robust (see Supplementary material online, *Table S11*). To address potential confounding factors, no instrumental variables for SHBG were associated with scoliosis or Marfan syndrome at genome-wide significance (see Supplementary material online, *Table S3*). After excluding BMI-related SNPs, the associations between SHBG and MVP remained significant in both two-sample and one-sample MR (see Supplementary material online, *Figure S4*). These findings suggest that the observed associations are unlikely to be driven by sample overlap or unmeasured confounding factors, further supporting the robustness of the MR results.

Table 2 Associations of SHBG and sex hormones with incident MVP

Exposure	HR (95% CI)	*P*
Crude model		
SHBG	1.524 (1.331, 1.745)	1.03×10^{-9}
Testosterone	1.196 (1.131, 1.263)	2.37×10^{-10}
Oestradiol[a]	0.743 (0.529, 1.042)	0.0853
Adjusted model[b]		
SHBG	1.440 (1.201, 1.726)	8.15×10^{-5}
Testosterone	1.037 (0.861, 1.250)	0.6992
Oestradiol[a]	0.987 (0.663, 1.470)	0.9504

In a cohort of 472 178 participants from the UK biobank with a median follow-up of 13.8 years, Cox proportional hazards models were employed to investigate the associations of serum levels of sex hormone-binding globulin (SHBG), testosterone, and oestradiol with the incidence of mitral valve prolapse (MVP).
CI, confidence interval; HR, hazard ratio; MVP, mitral valve prolapse; SHBG, sex hormone-binding globulin.
[a]Only 70 696 participants were detected with oestradiol in the total population and only were detected with oestradiol.
[b]Adjusted for age, sex, education, current smoking, current drinking, BMI, hypertension, diabetes mellitus, dyslipidaemia, and physical activity.

Discussion

In this study, we observed positive associations of serum SHBG concentration with the incidence and prognosis of MVP, and the impacts of SHBG on MVP did not vary by sex and other characteristics. Mendelian randomization analyses demonstrated a causal relationship between SHBG levels and MVP risk in both males and females. Our findings echoed the primary hypothesis and provided novel and important evidence regarding risk factors for MVP, offering new insights into the risk prediction and pathogenesis.

Previous studies showed that low SHBG levels were associated with metabolic disorders, including obesity, diabetes, and hyperlipidaemia.[13] As to cardiovascular diseases, although previous studies reported that SHBG concentrations are inversely related to the risk of coronary artery disease and stroke,[35,36] they may positively relate to the risk of atrial fibrillation/flutter, ventricular arrhythmias, bradycardia, venous thrombosis, mitral annular calcification, and ascending thoracic aortic calcification.[37-39] Our findings now add to the relationship of sex hormones with cardiovascular diseases, providing strong evidence on the positive association of SHBG with MVP. To do so, we leveraged from a large prospective cohort and employed multivariable models, sex-stratified analyses, and rigorous statistical methods to minimize the influence of confounding factors. In the MR analysis, we used BMI-adjusted SHBG GWAS data, applied more conventional and

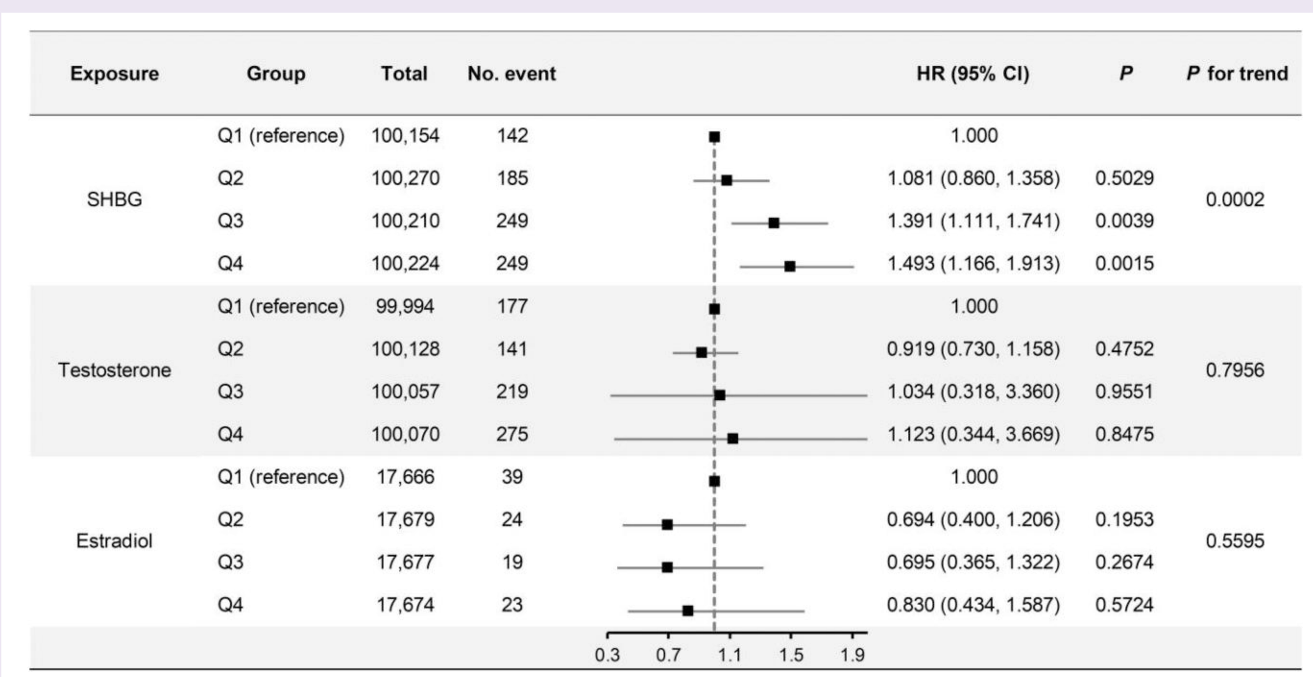


Figure 1 The serum concentrations of sex hormone-binding globulin and sex hormones were log-transformed before analysis. Quantile forms of sex hormone-binding globulin and sex hormones were used to evaluate their associations with mitral valve prolapse using Cox proportional hazards models, adjusted for age, education, current smoking, current alcohol consumption, body mass index, hypertension, diabetes, dyslipidaemia, and physical activity. CI, confidence interval; HR, hazard ratio; Q, quartile; SHBG, sex hormone-binding globulin; BMI, body mass index.

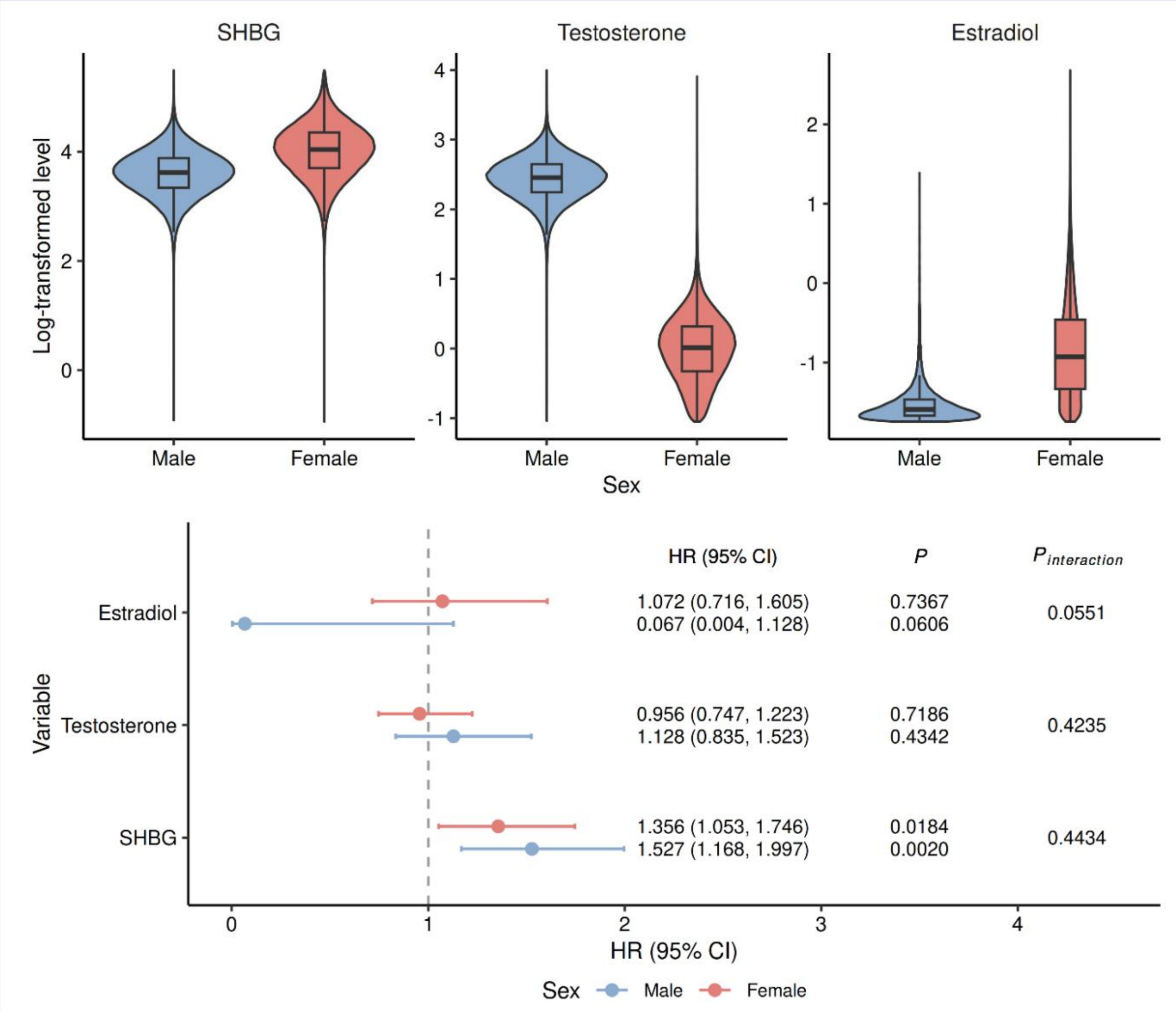


Figure 2 (*A*) Serum levels of sex hormones and sex hormone-binding globulin in males and females. Notably, only 70 696 participants were detected oestradiol. (*B*) Sex-stratified associations between sex hormone-binding globulin levels and incident mitral valve prolapse, estimated using Cox proportional hazards models adjusted for age, education, current smoking, current alcohol consumption, body mass index, hypertension, diabetes, dyslipidaemia, and physical activity. Multiplicative interaction terms were included in the models to assess effect modification by sex; the *P*-value for interaction indicates the statistical significance of the difference between groups. CI, confidence interval; HR, hazard ratio; SHBG, sex hormone-binding globulin; MVP, mitral valve prolapse; BMI, body mass index.

stringent filtering methods, excluded SNPs related to confounding factors, and conducted sex-stratified analyses. Overall, this study is the first to demonstrate an independent causal relationship between SHBG levels and MVP risk.

In addition, we observed that there was no significant sex differences in the effect of SHBG on the risk and adverse outcomes of MVP in our study, which was inconsistent with our initial hypothesis. Our findings suggest that elevated SHBG levels may influence the development and progression of MVP in both males and females in a similar manner and magnitude. However, this does not exclude the potential role of SHBG in contributing to sex differences in the prevalence and incidence of MVP. Since females typically have higher baseline levels of SHBG than males, prolonged exposure to elevated SHBG concentrations may exert a cumulative effect, thereby influencing the development of MVP in women.

Our findings highlight the potential value of SHBG as a biomarker for disease risk prediction, clinical risk stratification, and as a possible target for therapeutic intervention. MVP has traditionally been considered a congenital or degenerative disorder rather than one driven by metabolic factors. Previous studies have indicated that SHBG is involved in metabolic pathways, and lower SHBG levels have typically been associated with disease risk.[40,41] Interestingly, our findings, which demonstrate a positive association between elevated SHBG levels and MVP risk, may suggest the involvement of non-metabolic pathways in MVP pathogenesis. This observation could provide novel insights into the underlying biological mechanisms of MVP. Extracellular matrix (ECM)

Table 3 **Association of SHBG and sex hormones with mortality and MACE among MVP patients**

Exposure	Outcome	HR (95% CI)	*P*
Crude model			
SHBG	All-cause mortality	1.810 (1.281, 2.558)	0.0008
	MACE	0.982 (0.719, 1.341)	0.9084
Testosterone	All-cause mortality	1.131 (0.987, 1.297)	0.0763
	MACE	1.102 (0.978, 1.242)	0.1111
Oestradiol	All-cause mortality	1.161 (1.018, 1.324)	0.6066
	MACE	0.378 (0.106, 1.344)	0.1326
Adjusted model[a]			
SHBG	All-cause mortality	2.223 (1.430, 3.457)	0.0004
	MACE	1.243 (0.842, 1.837)	0.2737
Testosterone	All-cause mortality	1.226 (0.790, 1.903)	0.3633
	MACE	0.839 (0.572, 1.231)	0.3700
Oestradiol	All-cause mortality	1.527 (0.389, 5.987)	0.5438
	MACE	1.503 (0.283, 7.982)	0.6324

In total 1253 mitral valve prolapse (MVP) participants, Cox proportional hazards models were employed to investigate the associations of serum levels of sex hormone-binding globulin (SHBG), testosterone, and oestradiol with the prognosis of MVP, including all-cause mortality, and major cardiovascular events (MACE).
HR, hazard ratio; CI, confidence interval; MACE, major cardiovascular events; MVP, mitral valve prolapse; SHBG, sex hormone-binding globulin.
[a]Adjusted for age, sex, education, current smoking, current drinking, BMI, hypertension, diabetes mellitus, dyslipidaemia, and physical activity.

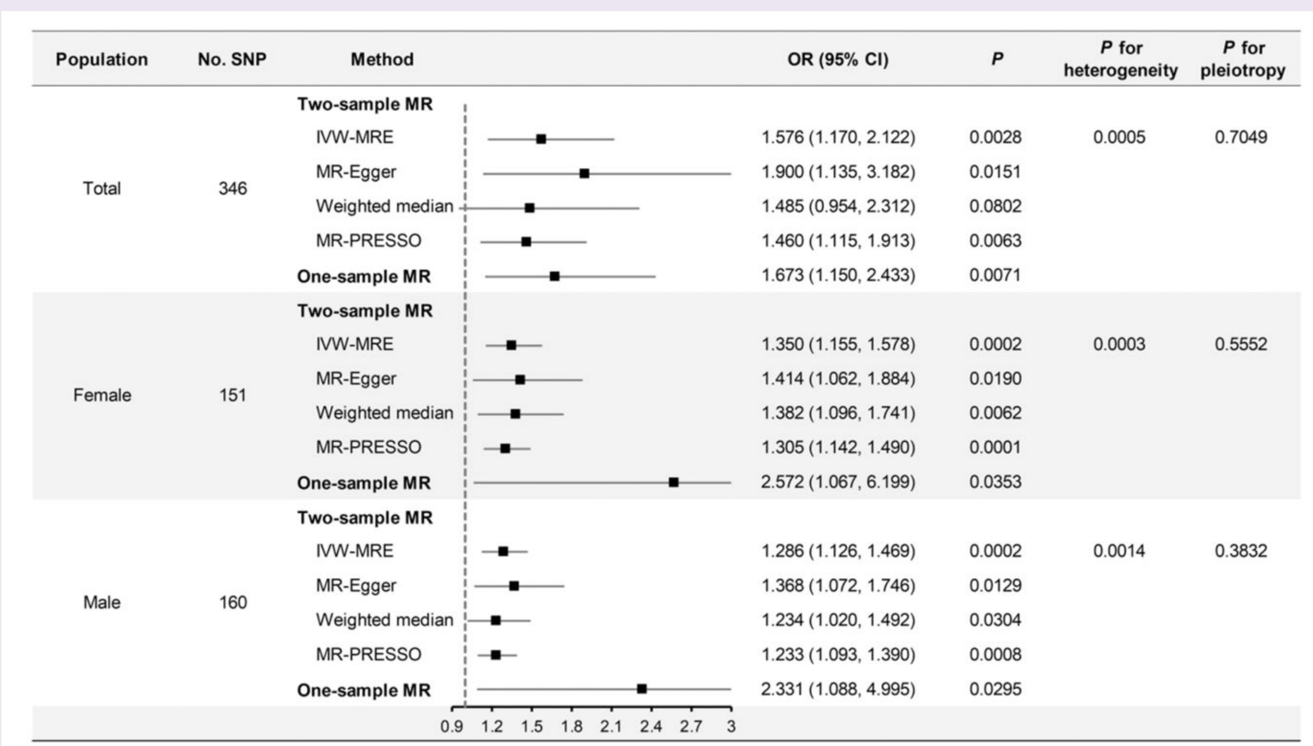


Figure 3 Casual association between sex hormone-binding globulin and mitral valve prolapse based on all eligible instrumental variables. The causal relationship between serum sex hormone-binding globulin and mitral valve prolapse was investigated using two-sample and one-sample Mendelian randomization. The causality was assessed among both male and female. Various methods were employed in the two-sample Mendelian randomization, with inverse variance weighting with multiplicative random effect as the primary evaluation metric, and Mendelian randomization Egger regression, weighted-median, and Mendelian randomization pleiotropy residual sum and outlier as the sensitivity analysis. CI, confidence interval; IVW-MRE, inverse variance weighting with multiplicative random effect; MVP, mitral valve prolapse; MR, Mendelian Randomization; MR-PRESSO, Mendelian randomization pleiotropy residual sum and outlier; MR-Egger, Mendelian randomization Egger regression; OR, Odds ratio; SNP, single nucleotide polymorphism.

remodelling driven by activated fibroblast-like valvular interstitial cells, along with inflammation and collagen degradation induced by infiltrating immune cells, constitutes the principal pathogenic mechanism of MVP.[2,42-44] Prior research has shown that SHBG is widely expressed in lymphocytes and is influenced by circulating SHBG concentrations. Circulating SHBG can initiate intracellular signalling in lymphocytes through membrane-bound receptors.[15] In rheumatic heart valve disease, SHBG may promote T-cell cytotoxicity via interaction with oestrogen receptors, potentially leading to damage of Type I collagen within the mitral valve.[16] Given that lymphocyte-mediated immune responses are important contributors to the development of mitral valve disease,[45,46] it is plausible that circulating SHBG could mediate mitral valve collagen damage by acting through oestrogen receptors on the surface of lymphocytes, particularly CD8+ T cells, thereby promoting the occurrence of MVP. Although we found that testosterone and oestradiol were not significantly associated with the incidence or prognosis of MVP, this does not exclude potential synergistic effects between SHBG and sex hormones on valvular pathophysiology, since SHBG is the major regulator of free testosterone and oestradiol levels. Oestrogen receptor alpha (ERα) signalling in T cells is known to be proinflammatory,[47] thus, oestradiol and SHBG may synergistically promote T cells mediated inflammatory processes within the mitral valve. Oestrogen receptor beta (ERβ) can modulate ECM homeostasis and tissue remodelling through the TGF-β/Smad signalling pathway.[48-51] Therefore, oestradiol and SHBG might influence valvular interstitial cell driven ECM remodelling via ERβ signalling, and disruptions in the dynamic balance between ERα and ERβ pathways could contribute to MVP pathogenesis. Previous studies have demonstrated that oestrogen and progesterone receptors are expressed in cardiac fibroblasts and can directly regulate ECM-related genes such as TGF-β3 and fibronectin, suggesting that sex hormone signalling plays a critical role in cardiac matrix remodelling. Given that SHBG modulates the bioavailability of sex hormones, altered SHBG levels may indirectly influence TGF-β mediated ECM remodelling in valvular interstitial cells.[52,53] Collectively, we hypothesis that SHBG may modulate the expression and balance of ER subtypes within immune cells and mitral valve tissue, either directly or through its regulation of sex hormone bioavailability. Although the precise molecular mechanisms remain to be elucidated, the SHBG-ER signalling axis represents a plausible mechanistic link between endocrine regulation and MVP. From a translational perspective, targeting SHBG or its downstream signalling interactions may offer a novel therapeutic avenue. Modulation of SHBG could potentially restore hormonal equilibrium and attenuate valvular changes. Future studies should characterize ER subtypes distribution and activity within MVP valve tissue and further define the interplay among SHBG, sex hormones, and ER mediated signalling to identify targets.

Strengths and limitations

Our study has several advantages. First, we utilized a large prospective cohort to demonstrate the association between SHBG and the incidence and prognosis of MVP. Second, leveraging large sample GWAS, we validated the genetic association between SHBG and MVP. Lastly, we strengthened our conclusions using sex-stratified data.

Our study also has several limitations. First, the limited number of individuals with oestrogen measurements in the UK Biobank may reduce the statistical power of analyses involving sex hormone levels. Second, the study population primarily consisted of older adults, with a lack of representation from younger adults and adolescents. Moreover, we were unable to examine the association between SHBG levels and specific MVP subtypes (e.g. Barlow disease and fibroelastic deficiency). In the MR analysis, GWAS summary statistics were derived exclusively from individuals of European ancestry. In the analysis of data from the UK Biobank, to maintain consistency with the GWAS data and minimize potential bias, non-White-European populations were excluded. This limited the generalizability of our findings to other ethnic groups. One key assumption of MR is that the genetic instruments affect the outcome only through the exposure (SHBG levels), and not via other biological pathways. Although we employed methods such as MR-Egger and MR-PRESSO to assess for potential horizontal pleiotropy, the possibility of residual pleiotropy cannot be entirely excluded. MVP diagnosis in this study was based on ICD-10 hospital admission codes. This approach ensures diagnostic specificity but limits our ability to distinguish disease severity, thereby introducing potential bias and limiting the generalizability of our findings to the broader MVP spectrum. While it is plausible that moderate-to-severe cases are more likely to be hospitalized and thus coded, it is also possible that mild cases diagnosed during routine clinical evaluations were included. Therefore, our cohort likely represents a clinically diagnosed subset of MVP rather than all cases in the community. Future studies in younger community-based populations, in non-European cohorts, and in samples with echocardiography-confirmed MVP diagnoses are warranted to further elucidate the relationship between SHBG and MVP.

Conclusion

This study demonstrated that elevated plasma SHBG levels were associated with the incidence and adverse outcomes (all-cause mortality) of MVP in a European population. We have thus identified a novel risk factor and prognostic predictor for MVP, providing clues for future investigations into the sex difference of MVP incidence, biological mechanisms underlying pathogenesis and clinical risk stratification. Further research is needed to confirm these findings in other population and to explore the biological mechanism.

Supplementary material

Supplementary material is available at *European Heart Journal—Quality of Care and Clinical Outcomes* online.

Author contributions

Z.F., Q.L., and G.S. contributed to the conception or design of the work. Z.F., Z.L., and K.L. contributed to the data acquisition for the work. Z.F., Q.L., Z.L., X.G., and X.L. contributed to the analysis and interpretation of data for the work. Z.F., Q.L., and Z.L. drafted the manuscript. Z.F., Q.L., Z.L., and G.S. critically revised the manuscript.

Acknowledgements

We thank all participants and investigators in the UK biobank for publicly available data.

Funding

This study was supported by the National Natural Science Foundation of China (Grant No. 82070404), Beijing Hospitals Authority's Ascent Plan (Grant No. DFL20240604), the Joint Funds for the innovation of science and Technology, Fujian province (Grant No. 2024Y96010029), the Youth Top Talent Project of Fujian Provincial Foal Eagle Program, Fujian Research and Training Grants for Young and Middle-aged Leaders in Healthcare.

Conflict of interest: None declared.

Data availability

The UK Biobank data that support the findings of this study can be accessed by researchers on application (https://www.ukbiobank.ac.uk/register-apply/).

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